

SELECT Query

```
SELECT col1, col2
FROM table
JOIN table2 ON table1.col = table2.col
WHERE condition
GROUP BY column_name
HAVING condition
ORDER BY col1 ASC|DESC;
```

SELECT Keywords

DISTINCT: Removes duplicate results	SELECT DISTINCT product_name FROM product;
BETWEEN: Matches a value between two other values (inclusive)	SELECT product_name FROM product WHERE price BETWEEN 50 AND 100;
IN: Matches to any of the values in a list	SELECT product_name FROM product WHERE category IN ('Electronics', 'Furniture');
LIKE: Performs wildcard matches using _ or %	SELECT product_name FROM product WHERE product_name LIKE '%Desk%';

Joins

```
SELECT t1.*, t2.*
FROM t1
join_type t2 ON t1.col = t2.col;
```

Table 1	Table 2
A	A
B	B
C	D

INNER JOIN: show all matching records in both tables.

A	A
B	B

LEFT JOIN: show all records from left table, and any matching records from right table.

A	A
B	B
C	

RIGHT JOIN: show all records from right table, and any matching records from left table.

A	A
B	B
	D

FULL JOIN: show all records from both tables, whether there is a match or not.

A	A
B	B
C	
	D

CASE Statement

Simple Case	CASE name WHEN 'John' THEN 'Name John' WHEN 'Steve' THEN 'Name Steve' ELSE 'Unknown' END
Searched Case	CASE WHEN name='John' THEN 'Name John' WHEN name='Steve' THEN 'Name Steve' ELSE 'Unknown' END

Common Table Expression

```
WITH queryname AS (  
    SELECT col1, col2  
    FROM firsttable)  
SELECT col1,col2..  
FROM queryname...;
```

Modifying Data

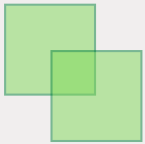
Insert	INSERT INTO tablename (col1, col2...) VALUES (val1, val2);
Insert from a Table	INSERT INTO tablename (col1, col2...) SELECT col1, col2...
Insert Multiple Rows	INSERT ALL INTO tablename (col1, col2) VALUES (valA1, valB1) INTO tablename (col1, col2) VALUES (valA2, valB2) SELECT * FROM dual;
Update	UPDATE tablename SET col1 = val1 WHERE condition;
Update with a Join	UPDATE t SET col1 = val1 FROM tablename t INNER JOIN table x ON t.id = x.tid WHERE condition;
Delete	DELETE FROM tablename WHERE condition;

Indexes

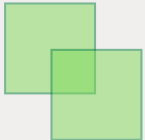
Create Index	CREATE INDEX indexname ON tablename (cols);
Drop Index	DROP INDEX indexname;

Set Operators

UNION: Shows unique rows from two result sets.



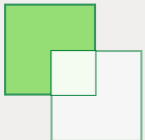
UNION ALL: Shows all rows from two result sets.



INTERSECT: Shows rows that exist in both result sets.



EXCEPT: Shows rows that exist in the first result set but not the second.



Aggregate Functions

- SUM: Finds a total of the numbers provided
- COUNT: Finds the number of records
- AVG: Finds the average of the numbers provided
- MIN: Finds the lowest of the numbers provided
- MAX: Finds the highest of the numbers provided

Common Functions

- LENGTH(string): Returns the length of the provided string
- INSTR(string, substring, [start_position], [occurrence]): Returns the position of the substring within the specified string.
- TO_CHAR(input_value, [fmt_mask], [nls_param]): Converts a date or a number to a string
- TO_DATE(charvalue, [fmt_mask], [nls_date_lang]): Converts a string to a date value.
- TO_NUMBER(input_value, [fmt_mask], [nls_param]): Converts a string value to a number.
- ADD_MONTHS(input_date, num_months): Adds a number of months to a specified date.
- SYSDATE: Returns the current date, including time.
- CEIL(input_val): Returns the smallest integer greater than the provided number.
- FLOOR(input_val): Returns the largest integer less than the provided number.
- ROUND(input_val, round_to): Rounds a number to a specified number of decimal places.
- TRUNC(input_value, dec_or_fmt): Truncates a number or date to a number of decimals or format.
- REPLACE(whole_string, string_to_replace, [replacement_string]): Replaces one string inside the whole string with another string.
- SUBSTR(string, start_position, [length]): Returns part of a value, based on a position and length.

Create Table

Create Table	CREATE TABLE tablename (column_name data_type);
Create Table with Constraints	CREATE TABLE tablename (column_name data_type NOT NULL, CONSTRAINT pkeyname PRIMARY KEY (col), CONSTRAINT fkname FOREIGN KEY (col) REFERENCES other_table(col_in_other_table), CONSTRAINT ucname UNIQUE (col), CONSTRAINT ckname CHECK (conditions));
Create Temporary Table	CREATE GLOBAL TEMPORARY TABLE tablename (colname datatype) ON COMMIT DELETE ROWS;
Drop Table	DROP TABLE tablename;

Alter Table

Add Column	ALTER TABLE tablename ADD columnname datatype;
Drop Column	ALTER TABLE tablename DROP COLUMN columnname;
Modify Column	ALTER TABLE tablename MODIFY columnname newdatatype;
Rename Column	ALTER TABLE tablename RENAME COLUMN currentname TO newname;
Add Constraint	ALTER TABLE tablename ADD CONSTRAINT constraintname constrainttype (columns);
Drop Constraint	ALTER TABLE tablename DROP constraint_type constraintname;
Rename Table	sp_rename 'old_table_name', 'new_table_name';

Window/Analytic Functions

```
function_name ( arguments ) OVER (  
    [query_partition_clause]  
    [ORDER BY order_by_clause  
    [windowing_clause] ] )
```

Example using RANK, showing the student details and their rank according to the fees_paid, grouped by gender:

```
SELECT  
student_id, first_name, last_name, gender, fees_paid,  
RANK() OVER (  
    PARTITION BY gender ORDER BY fees_paid  
) AS rank_val  
FROM student;
```

Subqueries

Single Row	SELECT id, last_name, salary FROM employee WHERE salary = (SELECT MAX(salary) FROM employee);
Multi Row	SELECT id, last_name, salary FROM employee WHERE salary IN (SELECT salary FROM employee WHERE last_name LIKE 'C%');

Numeric

NUMBER (p, s)	Numeric data, with a precision of p and scale of s. Precision p ranges from 1 to 38, and scale ranges from -84 to 127 Range: -10^38 +1 through 10^38 - 1
BINARY_FLOAT	32-bit, single-precision floating-point number Range: from 1.17549E-38F to 3.40282E+38F
BINARY_DOUBLE	64-bit, double-precision floating-point number Range: From 2.22507485850720E-308 to 1.79769313486231E+308
INTEGER	Synonym for NUMBER(38) Range: -10^38 +1 through 10^38 - 1
FLOAT	Translated to NUMBER. Range: 1 to 126 binary digits, and up to 22 bytes
DECIMAL	Translated to NUMBER. Range: 1 to 126 binary digits, and up to 22 bytes
NUMERIC	Synonym for NUMBER
INT	Synonym for NUMBER(38)
SMALLINT	Synonym for NUMBER(38)
DOUBLE PRECISION	Synonym for FLOAT(126)
REAL	Synonym for FLOAT(63)

Date

DATE	Stores date and time values. From Jan 1, 4712 BC to Dec 31 9999 AD
TIMESTAMP [(p)]	A date value that stores time and fractional seconds. From Jan 1, 4712 BC to Dec 31 9999 AD. Fractional seconds (p) must be 0 to 9
TIMESTAMP [(p)] WITH TIME ZONE	A timestamp that stores data in the specified timezone From Jan 1, 4712 BC to Dec 31 9999 AD. Fractional seconds (p) must be 0 to 9
TIMESTAMP [(p)] WITH LOCAL TIME ZONE	A timestamp that stores data in your local timezone. From Jan 1, 4712 BC to Dec 31 9999 AD. Fractional seconds (p) must be 0 to 9
INTERVAL YEAR [(p)] TO MONTH	Stores a period of time in years and months. Up to 2 digits in the year
INTERVAL DAY [(p)] TO SECOND	Stores a period of time in days, hours, minutes, and seconds. Between 0 and 9 digits in the day, between 0 and 9 digits fractional seconds

Character

CHAR (s)	Fixed-length character with a length of s. Up to 2,000 bytes.
NCHAR (s)	Fixed length national character with a length of s. Up to 2,000 bytes.
VARCHAR	Not implemented in Oracle.
VARCHAR2 (s)	Variable-length character string with a maximum length of s bytes. Up to 32,767 bytes
NVARCHAR2 (s)	Variable-length national character string with a maximum length of s bytes. Up to 32,767 bytes
LONG	Variable-length character string. Larger than VARCHAR2. Deprecated. Up to 2GB
RAW (s)	Raw binary data of length size. Up to 32,767 bytes
LONG RAW	Raw binary data of variable length. Deprecated. Up to 2 GB
CHARACTER VARYING CHARACTER	Synonym for VARCHAR2 Synonym for CHAR
BLOB	Stores large unstructured binary objects. Recommended for JSON fields before Oracle 21c Up to 8 TB to 128 TB (4 GB -1 * (database block size))
CLOB	Stores large single-byte and multi-byte objects. Up to 8 TB to 128 TB (4 GB -1 * (database block size))
NCLOB	Stores Unicode data. Up to 8 TB to 128 TB (4 GB -1 * (database block size))
JSON	Data type to store JSON data. New in Oracle 21c

Other

BFILE	A file locator that points to a file on the server. Up to 4GB
ROWID	Fixed-length binary data for a database row ID. Up to 10 bytes
UROWID	Universal RowID. Up to 4,000 bytes